3.5 半无限大物体的非稳态导热

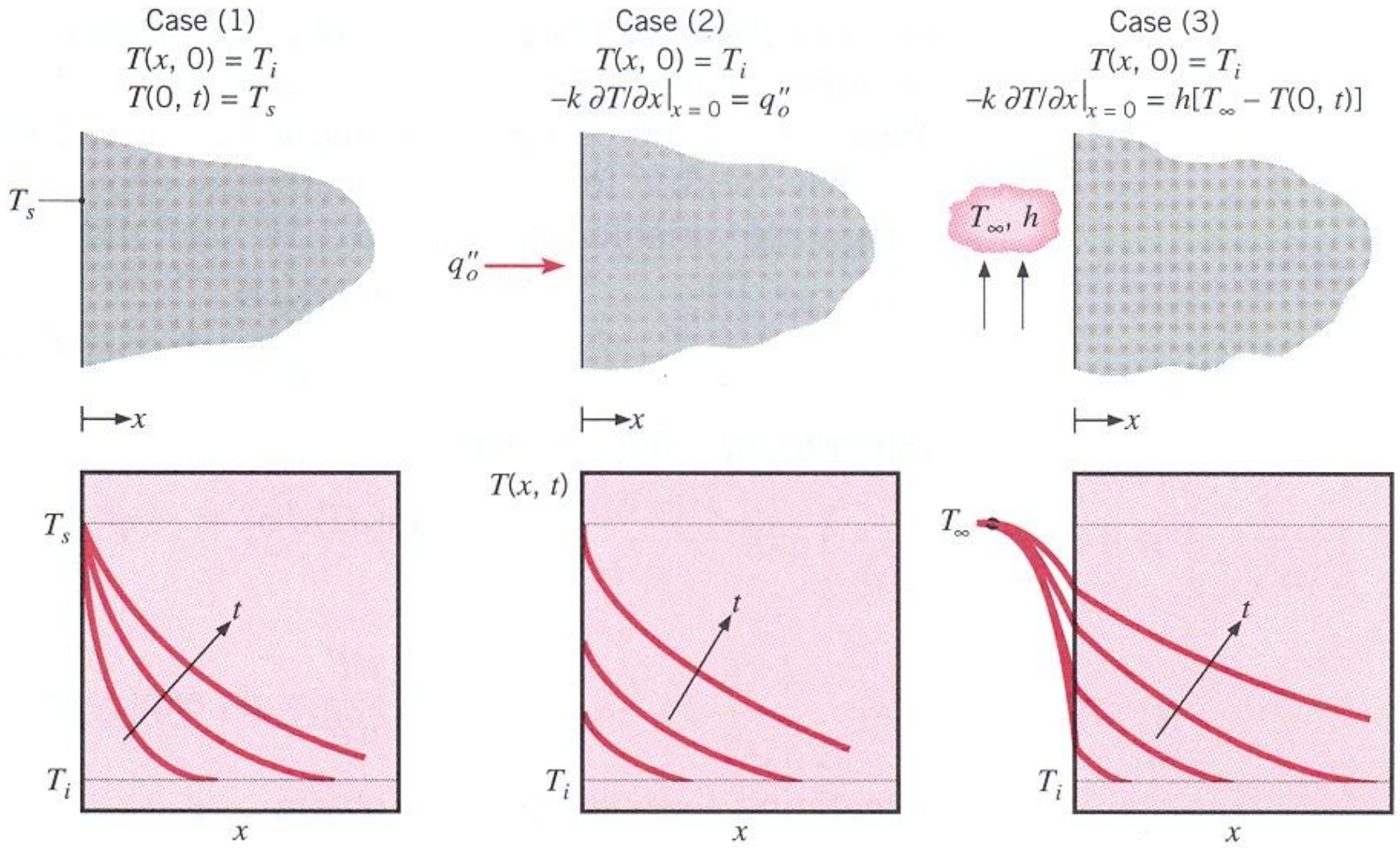


FIGURE 5.7 Transient temperature distributions in a semi-infinite solid for three surface conditions: constant surface temperature, constant surface heat flux, and surface convection.

问题的描述

一半无限大物体的初始温度为t_0，边界处的温度突然升高到t_s，物体的物性为常数，求物体内的温度分布？

导热微分方程与边界条件

$$\frac{\partial^2 t}{\partial x^2}=\frac{1}{a}\frac{\partial t}{\partial \tau}$$

$$t(x,0)=t_0$$

$$t(0,\tau)=t_s$$

$$t(\infty,\tau)=t_0$$

分析：

如果偏微分方程能够转化为常微分方程，问题将大大简化。

求解－相似变量法

定义如下组合变量：$\eta = \dfrac{x}{\sqrt{4a\tau}}$

$$\frac{\partial t}{\partial x} = \frac{dt}{d\eta}\cdot\frac{\partial \eta}{\partial x} = \frac{1}{\sqrt{4a\tau}}\cdot\frac{dt}{d\eta}$$

$$\frac{\partial^2 t}{\partial x^2} = \frac{d}{d\eta}\left(\frac{\partial t}{\partial x}\right)\cdot\frac{\partial \eta}{\partial x} = \frac{1}{4a\tau}\cdot\frac{d^2 t}{d\eta^2}$$

$$\frac{\partial t}{\partial \tau} = \frac{dt}{d\eta}\cdot\frac{\partial \eta}{\partial \tau} = -\frac{x}{2\tau\sqrt{4a\tau}}\cdot\frac{dt}{d\eta}$$

$$\begin{cases} \dfrac{d^2t}{d\eta^2} = -2\eta\dfrac{dt}{d\eta} \\ t(\eta = 0) = t_s \\ t(\eta \to \infty) = t_0 \end{cases}$$

$$\Rightarrow \quad \frac{d(dt/d\eta)}{dt/d\eta} = -2\eta d\eta$$

$$\Downarrow$$

$$\ln(dt/d\eta) = -\eta^2 + c'$$

$$\Downarrow$$

$$t = c_1 \int_0^{\eta} \exp(-\eta^2) d\eta + c_2$$

$$t(\eta = 0) = t_s \quad \Rightarrow \quad c_2 = t_s$$

$$t(\eta \to \infty) = t_0 \quad \Rightarrow \quad c_1 = \frac{2(t_0 - t_s)}{\sqrt{\pi}}$$

$$\int_0^{\infty} \exp(-\eta^2) d\eta = \frac{\sqrt{\pi}}{2}$$

$$\theta = \frac{t - t_s}{t_0 - t_s} = \frac{2}{\sqrt{\pi}} \int_0^{\eta} \exp(-\eta^2) d\eta = erf(\eta)$$

讨　论

- 分析解表明，边界上温度的突然上升，在无穷远处马上会有温度响应；
- 该谬误是基于傅立叶定律的导热微分方程所导致的。因此，可以说，虽然傅立叶定律有广泛的应用，但在某些场合可能会不成立；
- 是不是上述解不正确呢？

误差函数衰减得很快，只要η>3.2，当地的温度响应就可以忽略不计了

erf(3.2) = 0.99999

erf(2.0) = 0.9953

$$\eta = \frac{x}{\sqrt{4a\tau}} = 2.0 \Longrightarrow \text{若} \quad d \geq 4\sqrt{a\tau}$$

厚度为2d的无限大平板可用半无限大物体的解分析。

由温度分布，可以计算出壁面热流

$$q_w = -\lambda \left.\frac{\partial t}{\partial x}\right|_{x=0} = \lambda \frac{t_s - t_0}{\sqrt{\pi a \tau}} \exp\left(-\frac{x^2}{4a\tau}\right)\Bigg|_{x=0} = \frac{\lambda\left(t_s - t_0\right)}{\sqrt{\pi a \tau}}$$

总传热量

$$\Phi = A\int_0^{\tau} q_w d\tau = 2A\sqrt{\frac{\tau}{\pi}\rho c\lambda}\left(t_s - t_0\right)$$

$\sqrt{\rho c\lambda}$ 称为吸热系数，表示向与其接触的高温物体吸热的能力

等热流边界（constant surface heat flux）

$$t(x,\tau)-t_0=\frac{2q_0\left(a\tau/\pi\right)^{1/2}}{\lambda}\exp\left(-\frac{x^2}{4a\tau}\right)-\frac{q_0 x}{\lambda}erfc\left(\frac{x}{\sqrt{4a\tau}}\right)$$

对流边界（surface convection）

$$\frac{t(x,\tau)-t_0}{t_f-t_0}=erfc\left(\frac{x}{\sqrt{4a\tau}}\right)-\left[\exp\left(\frac{hx}{\lambda}-\frac{h^2a\tau}{\lambda^2}\right)\right]\left[erfc\left(\frac{x}{\sqrt{4a\tau}}+\frac{\sqrt{h^2a\tau}}{\lambda}\right)\right]$$

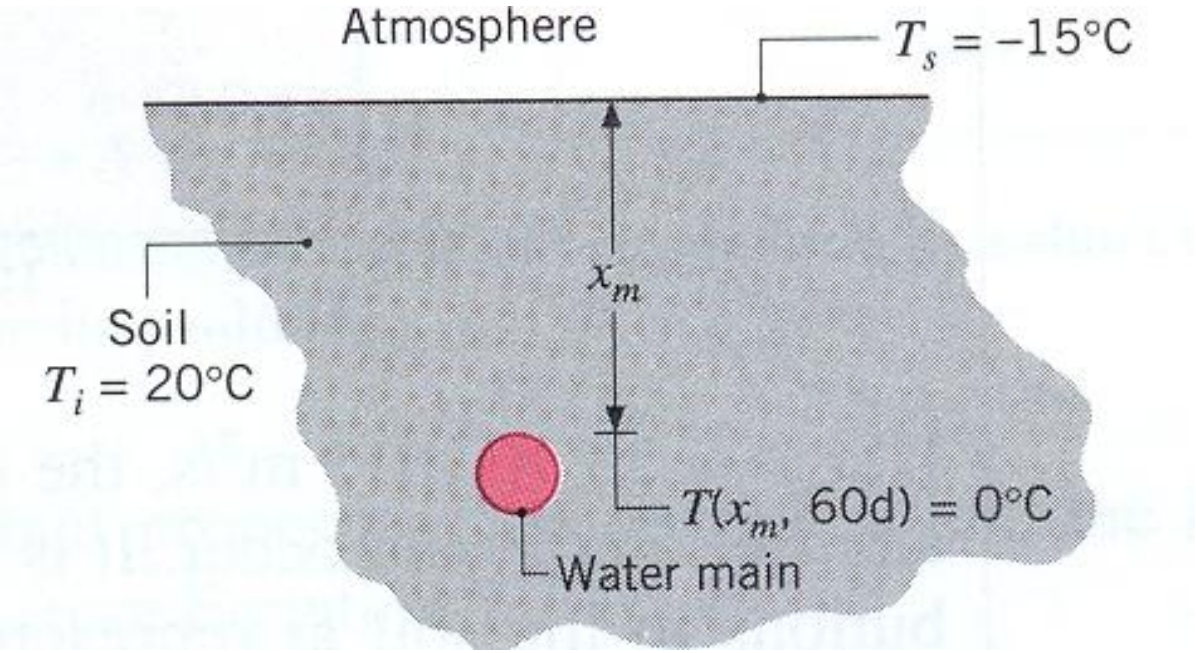
$$erfc(\eta)=1-erf(\eta)$$

问　题

半无限大物体的非稳态导热问题，是否也象无限大平板那样，存在正规加热阶段，为什么？

Example

In laying water mains, utilities must be concerned with the possibility of freezing during cold period. What minimum burial depth would you recommend to avoid freezing under conditions for which soil, initially at a uniform temperature of 20℃ , is subjected to a constant surface temperature of -15 ℃ for 60 days?



Assumptions:

- **One-Dimensional conduction in x**
- **Soil is a semi-infinite medium**
- **Constant properties**

Properties:

Soil(300 K): ρ=2050 kg/m², λ=0.52 W/mK

c=1840 J/kg, α=0.138×10^{-6} m²/s

Analysis:

$$\frac{T(x_m,\tau)-T_s}{T_i-T_s}=erf\left(\frac{x_m}{\sqrt{4a\tau}}\right)$$

$$\frac{0-(-15)}{20-(-15)}=0.429=erf\left(\frac{x_m}{\sqrt{4a\tau}}\right)$$

$$\left(\frac{x_m}{\sqrt{4a\tau}}\right)=0.40 \quad \Rightarrow \quad x_m = 0.68 \text{ m}$$

本章小结

- 非稳态导热与稳态导热的区别；特征；
- 集总参数法；Bi, Fo;
- 无限大平板问题的分析解；
- 正规加热阶段；
- 半无限大物体的非稳态导热；
- 傅立叶导热定律的适用性。

第四章 热传导问题的数值解法

4.1 数值解法的基本思想

将时间和空间坐标系中的连续物理量场用有限个离散点上的物理量的值的集合来代替；

由基本物理定律出发，建立离散点上的物理量的代数方程组，通过数值方法求解该方程组，获得所要求的物理量场。

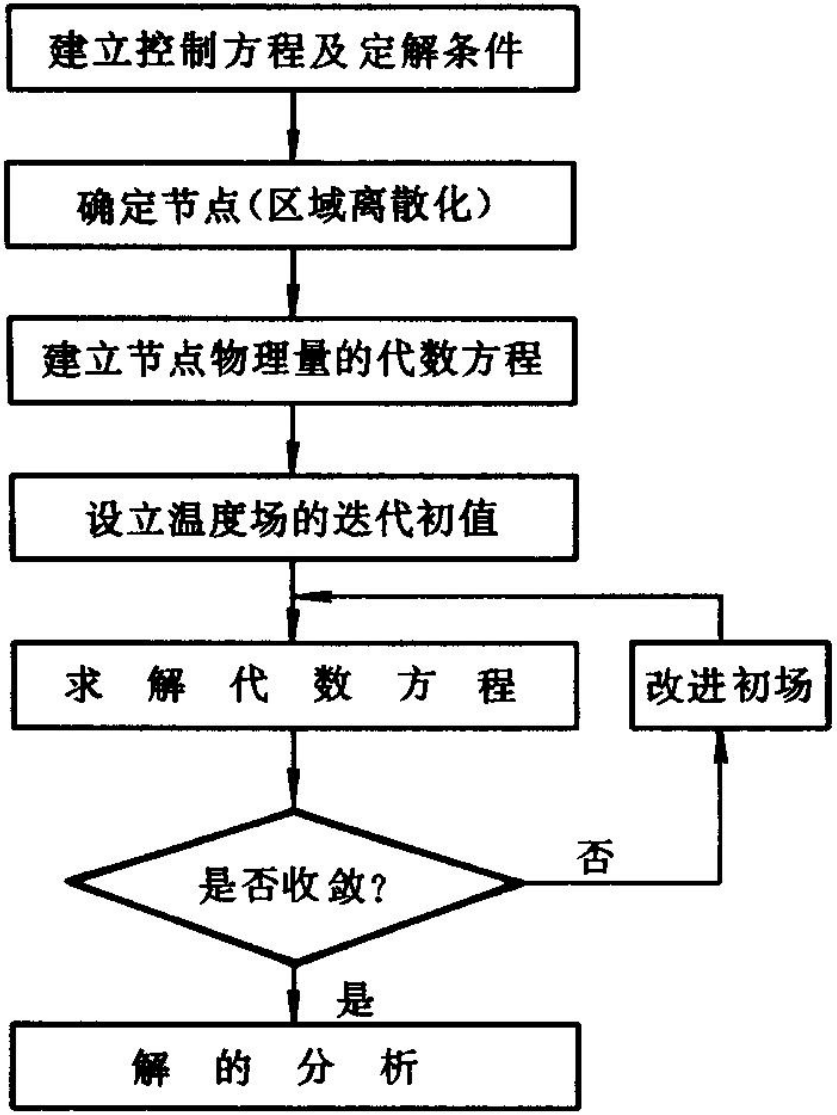


图 4-1　物理问题的数值求解过程

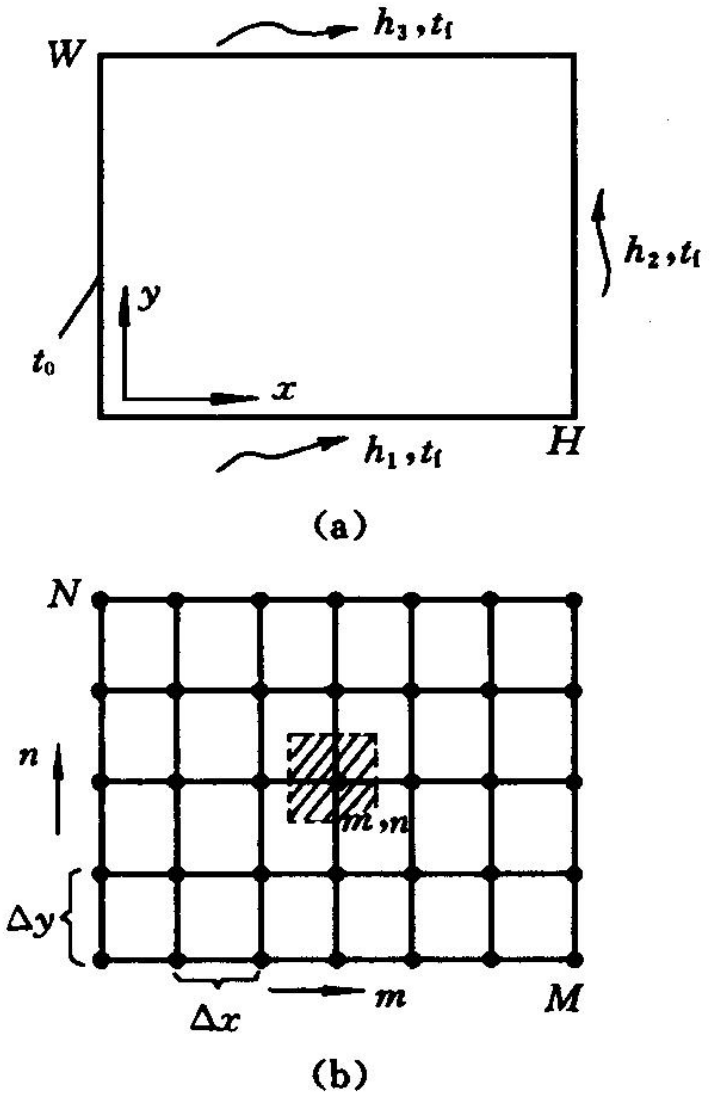
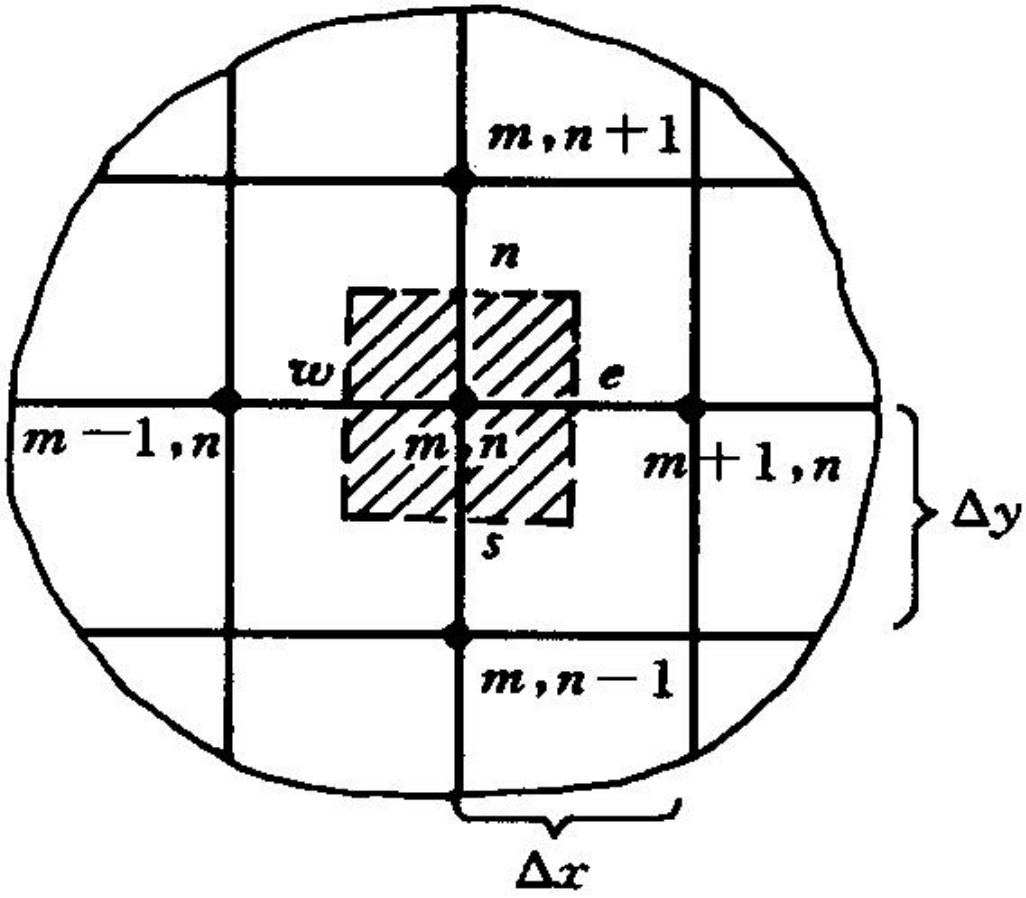


图 4-2　导热问题数值求解示例

4.2 节点离散方程的建立

以矩形域内的二维稳态导热为例，用平行于坐标轴的若干条线将区域划分为若干单元，交点为节点，两节点之间的距离为步长（两时间间隔称为时间步长）。

内部节点的离散方程



Taylor级数展开法

$$t_{m+1,n} = t_{m,n} + \left.\frac{\partial t}{\partial x}\right|_{m,n} \Delta x + \left.\frac{(\Delta x)^2}{2!}\frac{\partial^2 t}{\partial x^2}\right|_{m,n} + \left.\frac{(\Delta x)^3}{3!}\frac{\partial^3 t}{\partial x^3}\right|_{m,n} + \cdots$$

$$t_{m-1,n} = t_{m,n} - \left.\frac{\partial t}{\partial x}\right|_{m,n} \Delta x + \left.\frac{(\Delta x)^2}{2!}\frac{\partial^2 t}{\partial x^2}\right|_{m,n} - \left.\frac{(\Delta x)^3}{3!}\frac{\partial^3 t}{\partial x^3}\right|_{m,n} + \cdots$$

$$\left.\frac{\partial^2 t}{\partial x^2}\right|_{m,n} = \frac{t_{m-1,n} - 2t_{m,n} + t_{m+1,n}}{(\Delta x)^2}$$

$$\left.\frac{\partial^2 t}{\partial y^2}\right|_{m,n} = \frac{t_{m,n-1} - 2t_{m,n} + t_{m,n+1}}{(\Delta y)^2}$$

代入微分方程，得，

$$\frac{t_{m-1,n}-2t_{m,n}+t_{m+1,n}}{(\Delta x)^2}+\frac{t_{m,n-1}-2t_{m,n}+t_{m,n+1}}{(\Delta y)^2}=0$$

若$\Delta x=\Delta y$，则有，

$$t_{m,n}=\left(t_{m-1,n}+t_{m+1,n}+t_{m,n-1}+t_{m,n+1}\right)/4$$

稳态、常物性、无内热源的二维导热，当$\Delta x=\Delta y$时，任一内部节点的温度应等于其周围四个节点温度的算术平均值。按此推论，最高与最低温度一定在边界上（有内热源时不是如此），等温线要么为域内的封闭曲线，要么终止在边界上。

热平衡法

$$\lambda\Delta y\frac{t_{m-1,n}-t_{m,n}}{\Delta x}+\lambda\Delta y\frac{t_{m+1,n}-t_{m,n}}{\Delta x}$$

$$+\lambda\Delta x\frac{t_{m,n-1}-t_{m,n}}{\Delta y}+\lambda\Delta x\frac{t_{m,n+1}-t_{m,n}}{\Delta y}=0$$

若热导率为常数，且$\Delta x=\Delta y$，则，

$$t_{m,n}=\left(t_{m-1,n}+t_{m+1,n}+t_{m,n-1}+t_{m,n+1}\right)/4$$

边界节点的离散方程

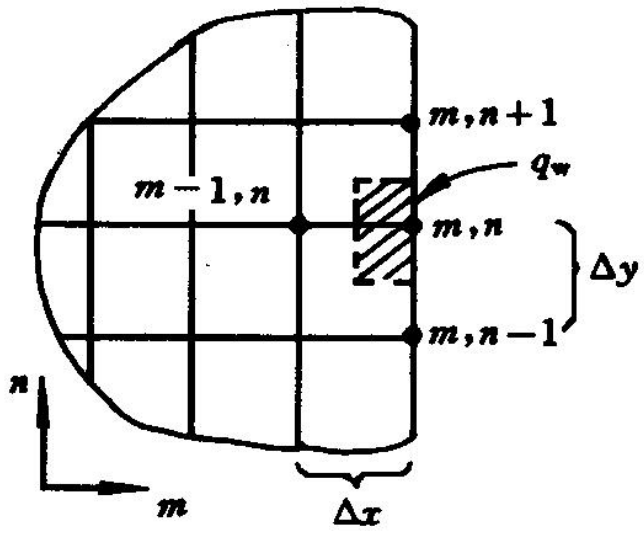


图 4-4　平直边界上的节点

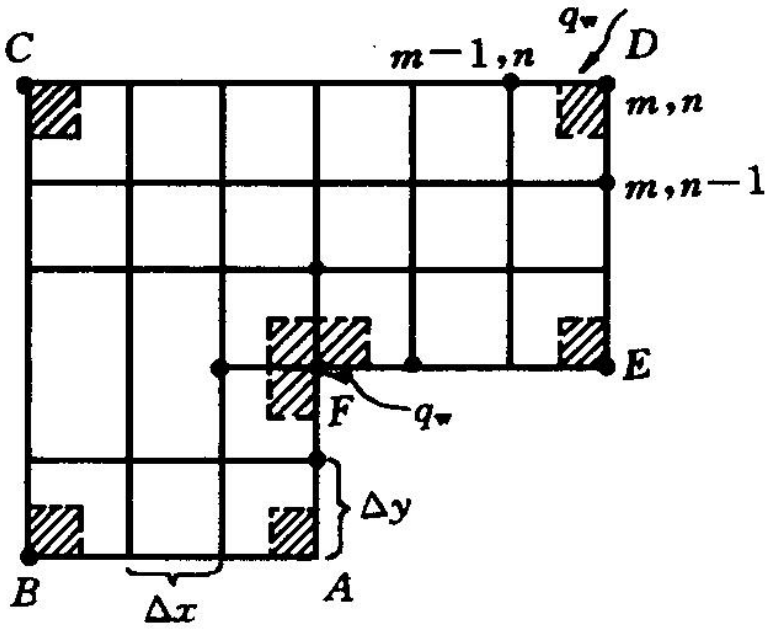
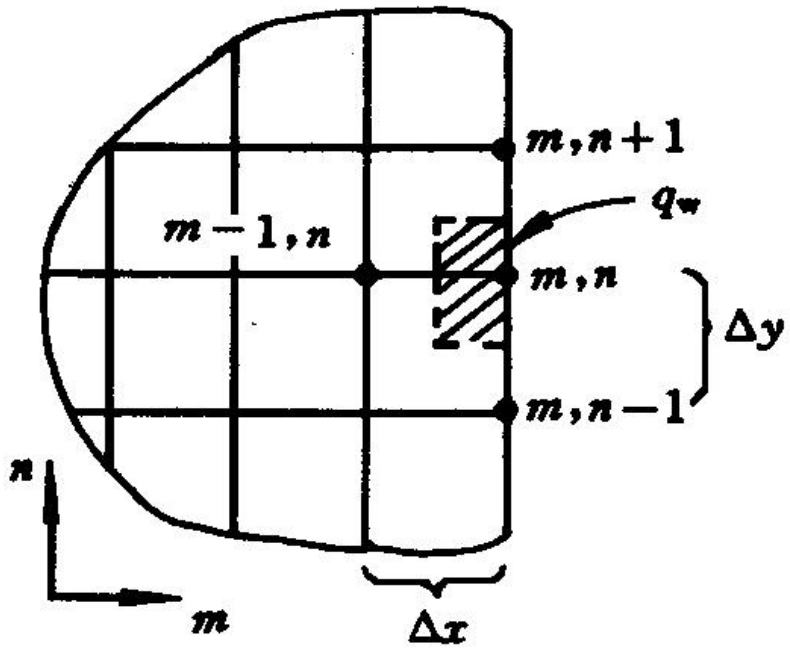


图 4-5　外部角点与内部角点

平直边界上的节点

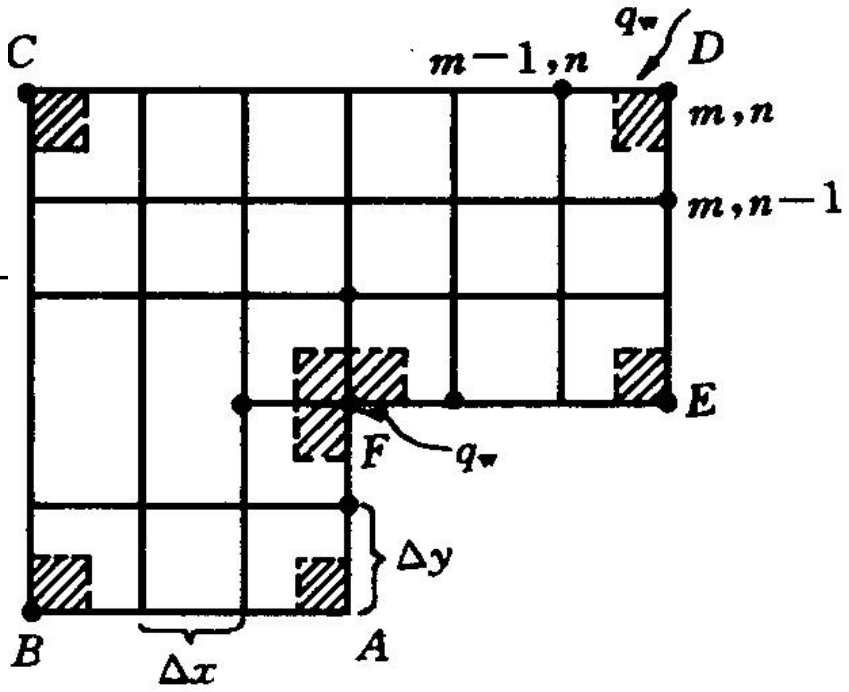


$$\lambda\Delta y\frac{t_{m-1,n}-t_{m,n}}{\Delta x}+\lambda\frac{\Delta x}{2}\frac{t_{m,n-1}-t_{m,n}}{\Delta y}+\lambda\frac{\Delta x}{2}\frac{t_{m,n+1}-t_{m,n}}{\Delta y}+q_w\Delta y=0$$

边界角点

$$\lambda\frac{\Delta x}{2}\frac{t_{m,n-1}-t_{m,n}}{\Delta y}+\lambda\frac{\Delta y}{2}\frac{t_{m-1,n}-t_{m,n}}{\Delta x}$$

$$+q_{w1}\frac{\Delta y}{2}+q_{w2}\frac{\Delta x}{2}=0$$



$$\lambda\Delta y\frac{t_{m-1,n}-t_{m,n}}{\Delta x}+\lambda\frac{\Delta y}{2}\frac{t_{m+1,n}-t_{m,n}}{\Delta x}+\lambda\frac{\Delta x}{2}\frac{t_{m,n-1}-t_{m,n}}{\Delta y}$$

$$+\lambda\Delta x\frac{t_{m,n+1}-t_{m,n}}{\Delta y}+q_w\frac{\Delta x+\Delta y}{2}=0$$

4.3 代数方程组的求解方法

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n = b_2 \\ \cdots\cdots\cdots \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \cdots + a_{nn}x_n = b_n \end{cases}$$

$$\begin{cases} x_1 = \dfrac{1}{a_{11}}\left(b_1 - a_{12}x_2 - a_{13}x_3 - \cdots - a_{1n}x_n\right) \\ x_2 = \dfrac{1}{a_{22}}\left(b_2 - a_{21}x_1 - a_{23}x_3 - \cdots - a_{2n}x_n\right) \\ \cdots\cdots\cdots \\ x_n = \dfrac{1}{a_{nn}}\left(b_n - a_{n1}x_1 - a_{n2}x_2 - \cdots - a_{n,n-1}x_{n-1}\right) \end{cases}$$

简单迭代法

设初值 $x_1^{(0)}, x_2^{(0)}, \cdots, x_n^{(0)}$

计算新值 $x_1^{(1)}, x_2^{(1)}, \cdots, x_n^{(1)}$

比较与原值的差 $x_i^k - x_i^{k-1} < \varepsilon$

迭代结束，如不满足，则重复运算，直至收敛。

高斯—赛德尔迭代法

用最新求出的x带入方程计算新值，而不是全部用原值

$$x_1^{k+1} = \frac{1}{a_{11}}\left(b_1 - a_{12}x_2^k - a_{13}x_3^k - \cdots - a_{1n}^k\right)$$

$$x_2^{k+1} = \frac{1}{a_{22}}\left(b_2 - a_{21}x_1^{k+1} - a_{23}x_3^k - \cdots - a_{2n}^k\right)$$

$$x_n^{k+1} = \frac{1}{a_{nn}}\left(b_2 - a_{n1}x_1^{k+1} - a_{n2}x_3^{k+1} - \cdots - a_{n,n-1}^{k+1}\right)$$

松弛法

$$x_i^k = \frac{1}{a_{ii}}\left(b_i - \sum_{i \neq j} a_{ij} x_j\right)$$

$$x_i^k = x_i^{k-1} + \frac{1}{a_{ii}}\left(b_i - \sum_{i \neq j} a_{ij} x_j\right) - x_i^{k-1}$$

$$x_i^{k'} = x_i^{k-1} + \alpha\left(\frac{1}{a_{ii}}\left(b_i - \sum_{i \neq j} a_{ij}x_j\right) - x_i^{k-1}\right)$$

$$= \left(1-\alpha\right)x_i^{k-1} + \alpha\left(\frac{1}{a_{ii}}\left(b_i - \sum_{i \neq j} a_{ij}x_j\right)\right)$$

$$x_i^{k'} = \left(1-\alpha\right)x_i^{k-1} + \alpha x_i^{k}$$

α>1,称为超松弛，α<1,称为欠松弛。采用松弛法，松弛因子选取得合适，既可加快收敛速度，也可保证解的收敛。

4.4 非稳态导热问题的数值解法

与稳态导热数值解的差别：

- 离散化包含空间域和时间域，对时间的差分形式不同导致不同的差分格式；
- 一维问题的方程为抛物型，与二维稳态导热问题的椭圆型方程不同。

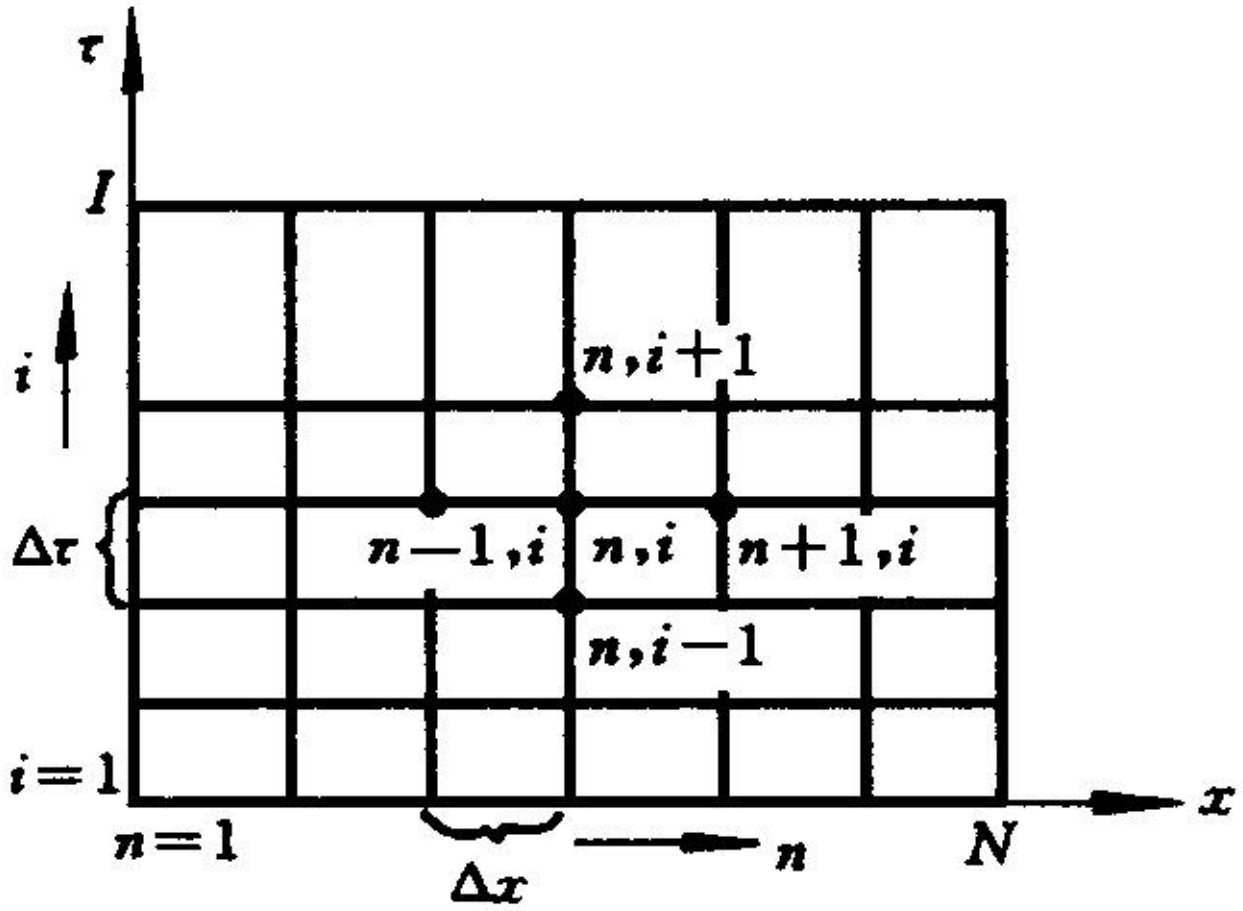
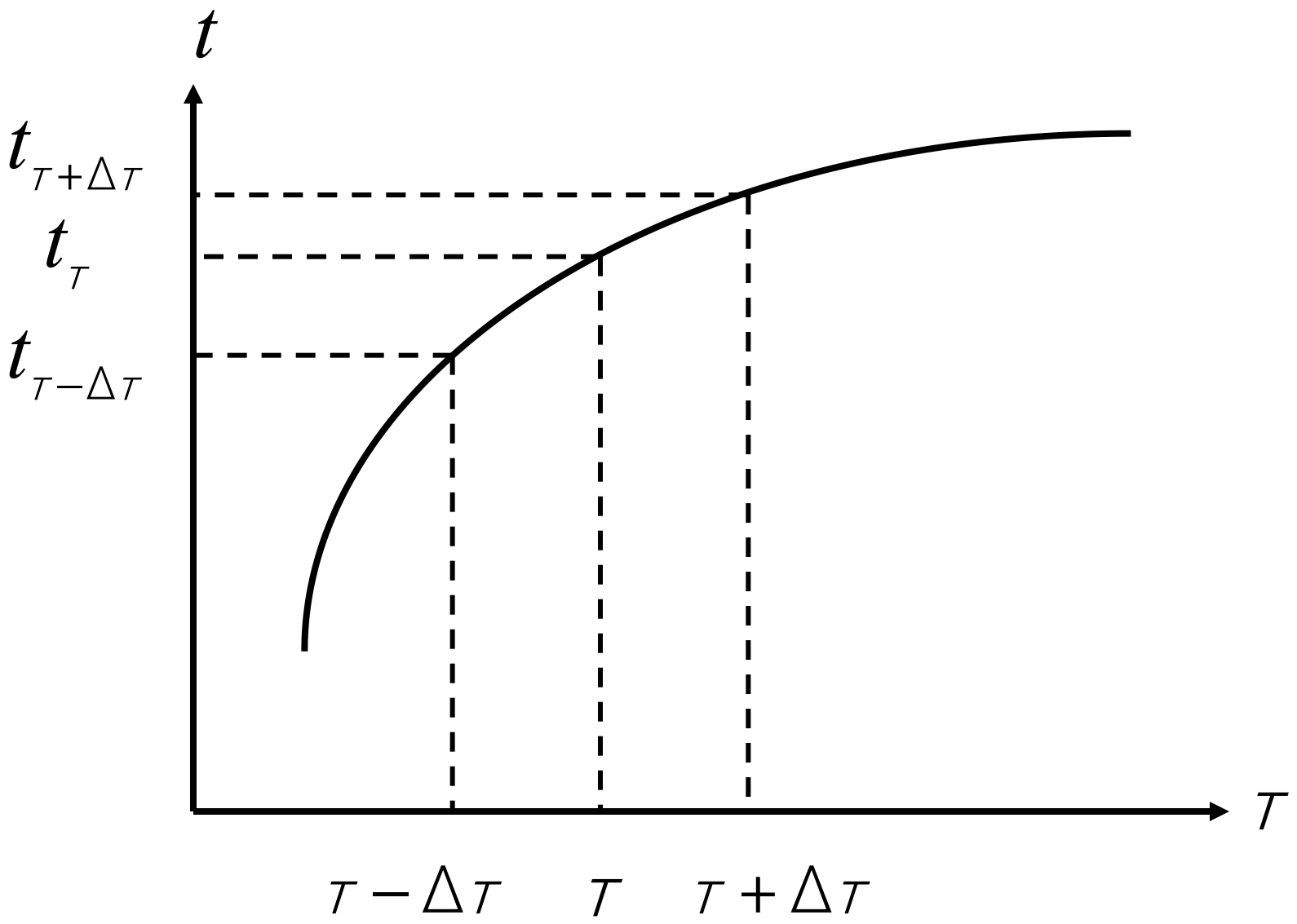


图 4-8　一维非稳态导热时间-空间区域的离散化


t
$t_{T+\Delta T}$
t_T
$t_{T-\Delta T}$
T
$T-\Delta T$
T
$T+\Delta T$

几种差分格式

向前差分

$$\left(\frac{\partial t}{\partial T}\right)_{T-\Delta T}=\frac{1}{\Delta T}\left(t_T-t_{T-\Delta T}\right)+O\left(\Delta T\right)$$

$$\left(\frac{\partial t}{\partial T}\right)_{T}=\frac{1}{\Delta T}\left(t_{T+\Delta T}-t_T\right)+O\left(\Delta T\right)$$

向后差分

$$\left(\frac{\partial t}{\partial T}\right)_{T}=\frac{1}{\Delta T}\left(t_T-t_{T-\Delta T}\right)+O\left(\Delta T\right)$$

$$\left(\frac{\partial t}{\partial T}\right)_{T+\Delta T}=\frac{1}{\Delta T}\left(t_{T+\Delta T}-t_T\right)+O\left(\Delta T\right)$$

中心差分 $$\left(\frac{\partial t}{\partial T}\right)_T = \frac{1}{2\Delta T}\left(t_{T+\Delta T} - t_{T-\Delta T}\right) + O\left(\left(\Delta T\right)^2\right)$$

Crank-Niclson格式

$$\frac{1}{2}\left(\left(\frac{\partial t}{\partial T}\right)_T + \left(\frac{\partial t}{\partial T}\right)_{T-\Delta T}\right) = \frac{1}{\Delta T}\left(t_T - t_{T-\Delta T}\right) + O\left(\left(\Delta T\right)^2\right)$$

通用差分格式

$$\sigma\left(\frac{\partial t}{\partial T}\right)_T + \left(1-\sigma\right)\left(\frac{\partial t}{\partial T}\right)_{T-\Delta T} = \frac{1}{\Delta T}\left(t_T - t_{T-\Delta T}\right) + \cdots$$

一维非稳态导热数值解的差分格式

微分方程 $$\frac{\partial^2 t}{\partial x^2}=\frac{1}{a}\frac{\partial t}{\partial T}$$

*i*节点 $$\left(\frac{\partial^2 t}{\partial x^2}\right)^i_n=\frac{1}{a}\left(\frac{\partial t}{\partial T}\right)^i_n$$

其中 $$\left(\frac{\partial^2 t}{\partial x^2}\right)_n^i = \frac{t_{n+1}^i - 2t_n^i + t_{n-1}^i}{(\Delta x)^2}$$

对时间的导数采用不同的差分格式，可获得不同的数值求解格式。

向前差分

$$\left(\frac{\partial t}{\partial \tau}\right)_n^{(i)} = \frac{1}{\Delta \tau}\left(t_n^{(i+1)} - t_n^{(i)}\right)$$

$$t_n^{(i+1)} = \frac{a\Delta \tau}{\Delta x^2}\left(t_{n+1}^{(i)} + t_{n-1}^{(i)}\right) + \left(1 - 2\frac{a\Delta \tau}{\Delta x^2}\right)t_n^{(i)}$$

向后差分

$$\left(\frac{\partial t}{\partial \tau}\right)_n^{(i+1)} = \frac{1}{\Delta \tau}\left(t_n^{(i+1)} - t_n^{(i)}\right)$$

$$\left(2 + \frac{(\Delta x)^2}{a\Delta \tau}\right) t_n^{(i+1)} = t_{n+1}^{(i+1)} + \frac{(\Delta x)^2}{a\Delta \tau} t_n^{(i)} + t_{n-1}^{(i+1)}$$

C-N格式

$$\frac{1}{2}\frac{t_{n+1}^{(i+1)}-2t_n^{(i+1)}+t_{n-1}^{(i+1)}}{(\Delta x)^2}+\frac{1}{2}\frac{t_{n+1}^{(i)}-2t_n^{(i)}+t_{n-1}^{(i)}}{(\Delta x)^2}=\frac{1}{a\Delta T}\left(t_n^{(i+1)}-t_n^{(i)}\right)$$

$$t_{n+1}^{(i+1)}-2\left(1+\frac{(\Delta x)^2}{a\Delta T}\right)t_n^{(i+1)}+t_{n-1}^{(i+1)}+t_{n+1}^{(i)}$$

$$+2\left(\frac{(\Delta x)^2}{a\Delta T}-1\right)t_n^{(i)}+t_{n-1}^{(i)}=0$$

通用隐式格式

$$(1-\sigma)\frac{t_{n+1}^{(i+1)}-2t_n^{(i+1)}+t_{n-1}^{(i+1)}}{(\Delta x)^2}+\sigma\frac{t_{n+1}^{(i)}-2t_n^{(i)}+t_{n-1}^{(i)}}{(\Delta x)^2}=\frac{1}{a\Delta T}\left(t_n^{(i+1)}-t_n^{(i)}\right)$$

正系数法则

任何地点、任何时刻，某点周围相邻点的温度的升高，必然导致该点温度的升高。

$$t_{m,n} = \left(t_{m-1,n} + t_{m+1,n} + t_{m,n-1} + t_{m,n+1}\right)/4$$

$$t_n^{(i+1)} = \frac{a\Delta\tau}{\Delta x^2}\left(t_{n+1}^{(i)} + t_{n-1}^{(i)}\right) + \left(1 - 2\frac{a\Delta\tau}{\Delta x^2}\right)t_n^{(i)}$$

要求：

$$1 - 2\frac{a\Delta\tau}{\Delta x^2} \geq 0$$

$$Fo_{\Delta x} \leq 0.5$$

作业：3-48,3-51,3-52，3-54,4-6